

Database Schema

Table 1: Departments

```
CREATE TABLE Departments (  
    DeptID INT PRIMARY KEY,  
    DeptName VARCHAR(50),  
    Location VARCHAR(50)  
);
```

Sample Data

```
INSERT INTO Departments VALUES  
(1, 'HR', 'Mumbai'),  
(2, 'IT', 'Pune'),  
(3, 'Finance', 'Delhi'),  
(4, 'Marketing', 'Bangalore');
```

Table 2: Employees

```
CREATE TABLE Employees (  
    EmpID INT PRIMARY KEY,  
    EmpName VARCHAR(50),  
    Age INT,  
    Gender VARCHAR(10),  
    Salary DECIMAL(10,2),  
    DeptID INT,  
    JoinDate DATE,  
    FOREIGN KEY (DeptID) REFERENCES Departments(DeptID)  
);
```

Sample Data

```
INSERT INTO Employees VALUES
(101, 'Amit', 25, 'Male', 35000, 2, '2022-01-15'),
(102, 'Priya', 30, 'Female', 45000, 1, '2021-05-20'),
(103, 'Rahul', 28, 'Male', 50000, 2, '2020-08-10'),
(104, 'Sneha', 35, 'Female', 60000, 3, '2019-03-25'),
(105, 'Vikas', 40, 'Male', 70000, 4, '2018-07-12'),
(106, 'Neha', 27, 'Female', 40000, 2, '2023-02-14'),
(107, 'Arjun', 32, 'Male', 55000, 3, '2021-11-05'),
(108, 'Pooja', 29, 'Female', 48000, 1, '2022-06-18'),
(109, 'Karan', 26, 'Male', 38000, 4, '2023-01-10'),
(110, 'Riya', 31, 'Female', 52000, 2, '2020-09-30');
```

Practice Questions

Section A: Basic Queries

Q1

Display all employee details.

Q2

Display only Employee Name and Salary.

Q3

Show employees whose salary is greater than 50000.

Q4

Show employees whose age is between 25 and 30.

Q5

Display employees who joined after 2021-01-01.

Q6

Display employees whose name starts with 'P'.

Q7

Display employees whose name ends with 'a'.

Q8

Display employees sorted by salary in descending order.

Q9

Display top 5 highest-paid employees.

Q10

Display all female employees.

Section B: DISTINCT

Q11

Display all distinct department IDs from Employees table.

Q12

Display all distinct genders.

Q13

Count distinct departments having employees.

Section C: Aggregate Functions

Q14

Find the total number of employees.

Q15

Find the maximum salary.

Q16

Find the minimum salary.

Q17

Find the average salary.

Q18

Find the total salary paid to all employees.

Q19

Find the average age of employees.

Q20

Find the number of female employees.

Section D: GROUP BY

Q21

Find the number of employees in each department.

Q22

Find the average salary department-wise.

Q23

Find the maximum salary department-wise.

Q24

Find the minimum salary department-wise.

Q25

Find the total salary department-wise.

Q26

Find the average age department-wise.

Section E: HAVING

Q27

Display departments having more than 2 employees.

Q28

Display departments whose average salary is greater than 45000.

Q29

Display departments where total salary exceeds 100000.

Q30

Display departments having at least 2 female employees.

Section F: JOINS

Q31

Display employee name along with department name.

Q32

Display employee name, salary, and department location.

Q33

Display all employees working in the IT department.

Q34

Display employees working in Mumbai location departments.

Q35

Display department name and number of employees in each department.

Q36

Display department-wise average salary.

Q37

Display the employee with department name ordered by department name.

Q38

Display all employees and their department details using INNER JOIN.

Q39

Display departments that currently have employees.

Q40

Display employee names and department names where salary is greater than 50000.

Section G: Subqueries

Q41

Find employees earning more than the average salary.

Q42

Find employees earning the highest salary.

Q43

Find employees earning the second highest salary.

Q44

Find employees who work in the same department as Amit.

Q45

Find employees earning less than the average salary.

Q46

Find employees working in the department with the highest average salary.

Q47

Find departments having more employees than the average department size.

Q48

Find employees whose salary is greater than all employees in HR.

Q49

Find employees belonging to the department located in Pune.

Q50

Display employee(s) having the minimum salary.
